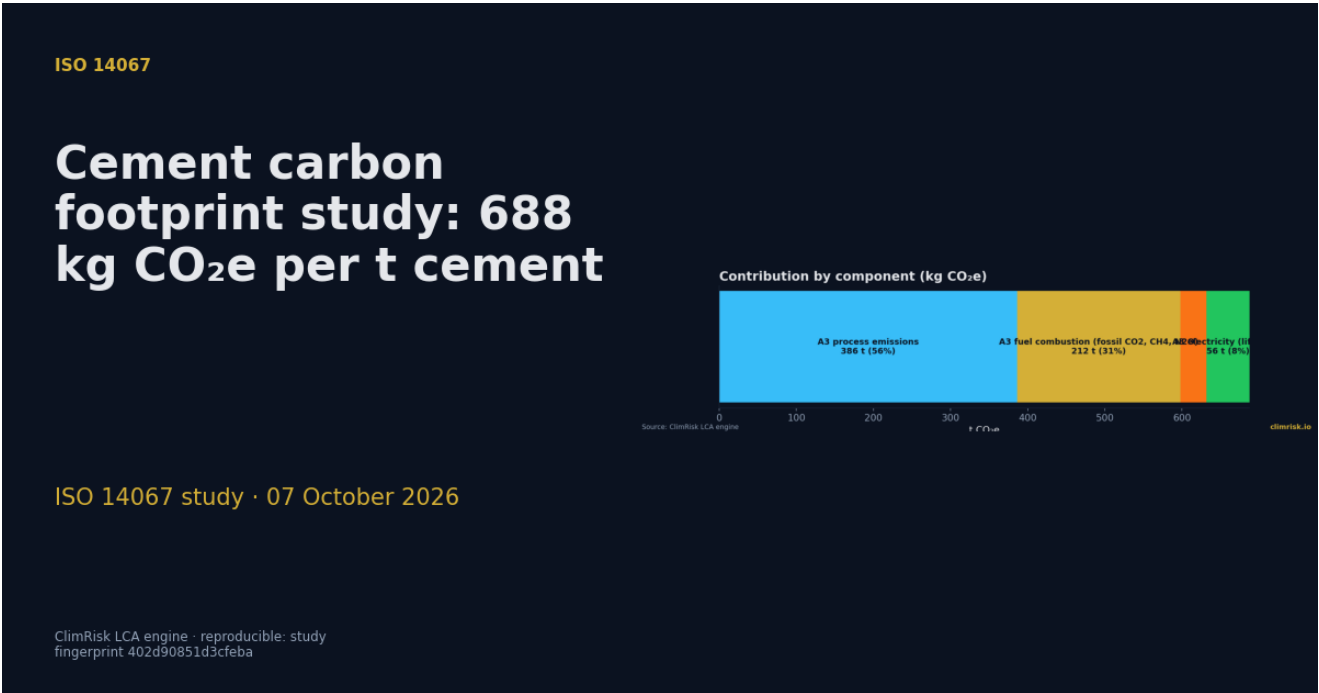


Cement carbon footprint study: 688 kg CO₂e per t cement

Partial CFP, cradle-to-gate (A1–A3). 688 kg CO₂e (P5–P95 573–822), IPCC AR6 WG1 GWP100, reference year 2025. 3 item(s) make up 80% of the footprint. A partial CFP: stages after the boundary are not included.

ClimRisk LCA engine · 2026-10-07



The study in six numbers

Goal, scope and results at a glance

688 kg

CO₂e per declared unit

Partial CFP, cradle-to-gate (A1–A3)

573–822

5th–95th percentile

Monte Carlo

3

items make 80% of the total

ISO 14067 §7 hotspot analysis

good

data quality, score 1.6

1 = best ... 5 = worst

44%

primary-data share

of emissions

2025

reference year

IPCC AR6 WG1 GWP100

Source: ClimRisk LCA engine; ISO 14067:2018

climrisk.io

Study summary computed from the inventory.

Calcination & raw-meal carbon (GCCA-CSI default 525 kg CO₂/t clinker (composition not provided)): 56% of the footprint

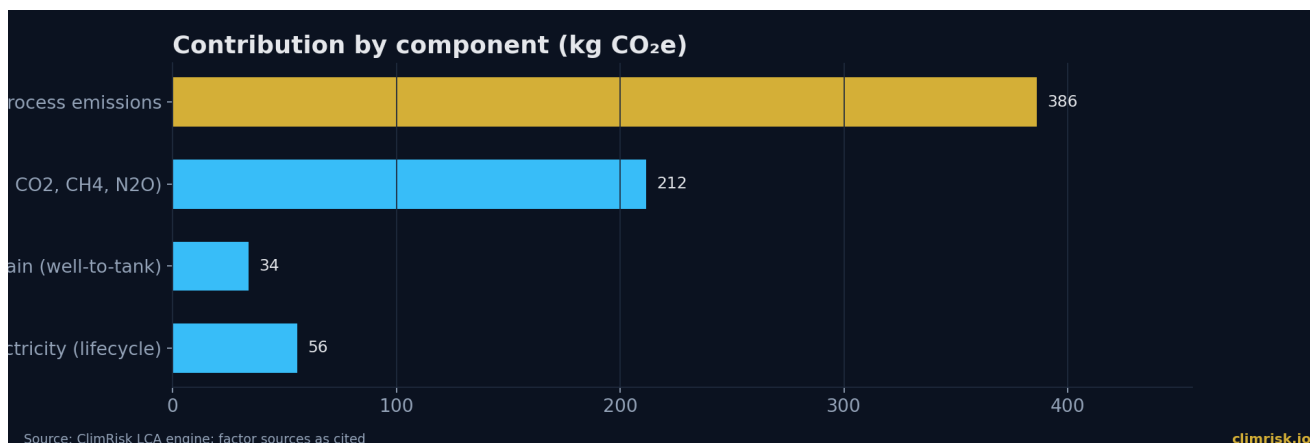
3 item(s) reach 80% cumulatively. The study is reproducible: the same inputs give fingerprint 402d90851d3cfeba.

Goal and scope

Intended application: not stated. Audience: not stated. A comparative assertion disclosed to the public requires a third-party critical review. [1]

Item	Value
Product	cement
Declared unit	1 t cement
System boundary	Partial CFP, cradle-to-gate (A1–A3)
Excluded stages	A4, B, C
Time period	reference year 2025
Geography	IND
GWP basis	IPCC AR6 WG1 GWP100
Allocation	none applied (single-output process)

Results



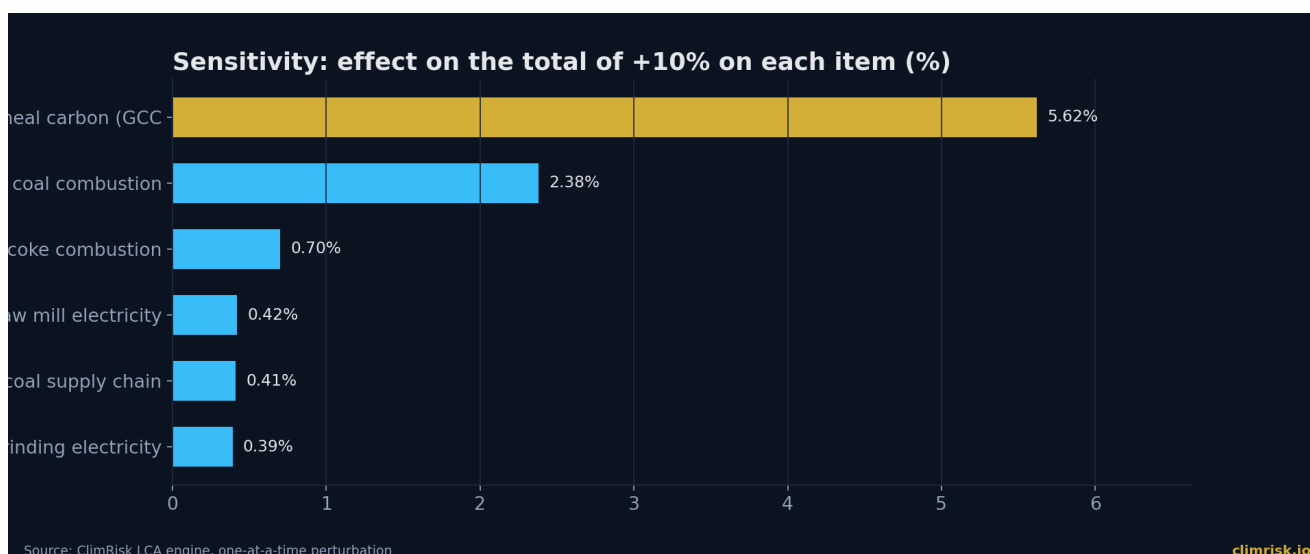
GHG type (ISO 14067 §6.3.5)	kg CO ₂ e
fossil ghg	687.61
biogenic co2 emissions	24.48
biogenic co2 removals	-24.48

Biogenic CO₂ emissions and removals are reported separately and are not included in the fossil total.

Interpretation: hotspots, sensitivity and methodological choices

Stage	Item	kg CO ₂ e	Share	Cumulative
A3	Calcination & raw-meal carbon (GCCA-CSI default 525 kg CO ₂ /t clinker (composition not provided))	386.2	56.2%	56.2%
A3	Other bituminous coal combustion	163.3	23.8%	79.9%
A3	Petroleum coke combustion	47.9	7.0%	86.9%

Items making up 80% of the footprint.



Methodological choice	Total kg CO ₂ e	Change vs selected
no biogenic uptake credit (biogenic CO ₂ counted as emitted)	712.1	+3.6%

Methodological choice	Total kg CO ₂ e	Change vs selected
gate emissions with the 2035 grid under NGFS Net Zero 2050	0.0	-6.8%
gate emissions with the 2035 grid under NGFS Current Policies	0.0	-2.5%

Data quality and cut-off

Overall data quality: good (score 1.6, 1 = best ... 5 = worst); primary data cover 44% of emissions.

Criterion	Score	Rating	Basis
Time-related coverage	1	good	oldest time-sensitive factor in use is 0 yr before the reference year 2025
Geographical coverage	1	good	plant-country grid factor used
Technological coverage	3	fair	44 % of emissions come from primary data (data-quality score ≤ 2)
Precision	3	fair	P5–P95 half-range is 18.3 % of the median
Completeness	2	good	0 declared exclusion(s); cut-off not assessed
Representativeness	2	good	mean of geographical and technological coverage
Consistency	1	good	one factor library, IPCC AR6 GWP100 and one boundary rule applied to every flow
Reproducibility	1	good	every input and factor is returned and hashed (fingerprint)
Sources of data	1	good	all factors cite a published table (see source list)
Uncertainty of the information	1	good	Monte Carlo over every flow, 1500 draws

Cut-off: not assessed. No excluded flows listed, so the mass and emission shares of the cut-off flows cannot be computed. Supply cutoff.excluded and cutoff.total_input_mass_t.

Review and communication

No third-party critical review has been recorded for this study.

Partial CFP, cradle-to-gate (A1–A3) of cement per 1 t cement: 688 kg CO₂e (P5–P95 573–822), IPCC AR6 GWP100, reference year 2025. Calculated with the ClimRisk LCA engine following ISO 14067:2018 methodology. No third-party critical review has been recorded for this study. Stages after the gate/customer boundary are not included.

Method and limitations

Calcination uses the GCCA-CSI default; provide clinker CaO/MgO for a Tier 2 result. Other primary solid biomass (biogenic) supply chain (no well-to-tank factor available) Study fingerprint 402d90851d3cfeba: the same inputs reproduce this result.

How ClimRisk measured this

- Inventory → named flows with emission factors from cited sources.
- Boundary and allocation → as declared above, per ISO 14067 §6.
- Uncertainty → Monte Carlo on factor and activity ranges; sensitivity by one-at-a-time perturbation.
- Reproducibility → a fingerprint of the inputs and factor library.

Sources

1. ISO 14067:2018 Greenhouse gases — Carbon footprint of products <https://www.iso.org/standard/71206.html>

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.